

BACHELOR PAPER

Term paper submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Engineering at the University of Applied Sciences Technikum Wien - Degree Program Electronic Engineering

Title

PID controlled levelling of a Nucleo-based teaching drone

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Wien, January 31, 2026

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Kurzfassung

Hier wird die Kurzfassung der Arbeit stehen.

Schlagworte: PID, Regelung, Drone, IMU, Kalman Filter

Abstract

The abstract of the thesis will be written here.

Keywords: PID, Control Loop, Drone, IMU, Kalman filter

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1 Introduction (2 Pages)

A short introduction into the background of the project and what can be expected by the work.

1.1 Goals

Goals set out to be achieved by the project.

1.2 Project description

Here an overview of the projects setup is given and project specific terms will be layed out.

2 Theoretical Background (5-10 Pages)

The theoretical and mathematical background of the controlling in general but specifically the PID controller and the methods for parameterization will be shown and equations are derived. The kalman filter used for combination of accelerometer and gyroscope data will also be thouroughly explained.

2.1 Control systems

2.2 PID Cotnrollers

2.3 Kalman filters

3 State of the Art (5 -10 Pages)

In this chapter the current state of drone flight controlling in scientific literature and industrial practice is analysed and key features will be highlighted.

3.1 History of Drones

3.2 State of flight controlling in literature

3.3 State of flight controlling in the industry

4 Implementation (10 Pages)

In this part the real implementation of the theory discussed early will be shown. code snippets and diagrams are used to aid in understanding exactly how the programm works and a reasoning for the way code is written is given. Tools used through out the process like the IDE or debugging tools like the logic analyzers will be mentioned.

4.1 Hardware Configuration

Giving an overview over the hardware its connection and the problems that arose during development aswell as how it effects the final implementation.

4.2 Control Algorithm

Here the real world implementation of the kalman filter and PID controller which were discussed prior in mathematical theory will be shown.

4.3 Software Architecture

In this section the implementation of the code will be explained and challenges, especially those specific to the plattform used, that arose during development will be mentioned.

5 Methodology (5 Pages)

Key test metrics and parameters from literature and industrial practice are identified and the testing implementation of will be presented.

5.1 Testing in literature and industry

The literature will be reviewed and common parameters used for comparing the feasibility of architectures will identified.

5.2 Testing approach

Explaining how this project was evaluated and why these steps were taken.

6 Results (3 Pages)

Here the achieved test results from the tests explained prior will be analysed and compared to the values from the literature. These finding will then be discussed and their meaning will be put into the projects context.

6.1 Acquired data

The acquired data will be presented.

6.2 Discussion

In this section the results generated will be evaluated and put into context.

List of Abbreviations

PID Proportional–Integral–Derivative